

Predicting Popular Recipes to Drive Site Traffic

Analysis & Recommendations

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The Project & Business Goal

Business Goal

Increase site-wide traffic and subscriptions (up to 40% lift from popular recipes).

The Problem

Difficult to manually predict which recipes will be popular.

The Request

Predict high-traffic recipes 80% of the time.

Our Work: A Data-Driven Approach

1

Data Validation

Cleaned and standardized the provided recipe data.

2

Exploratory Analysis

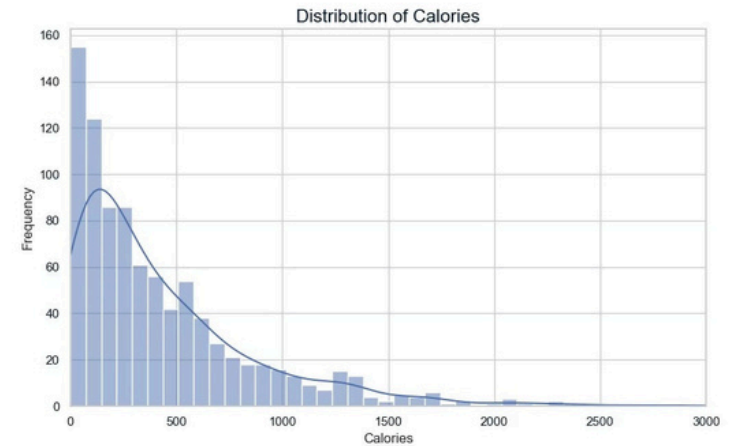
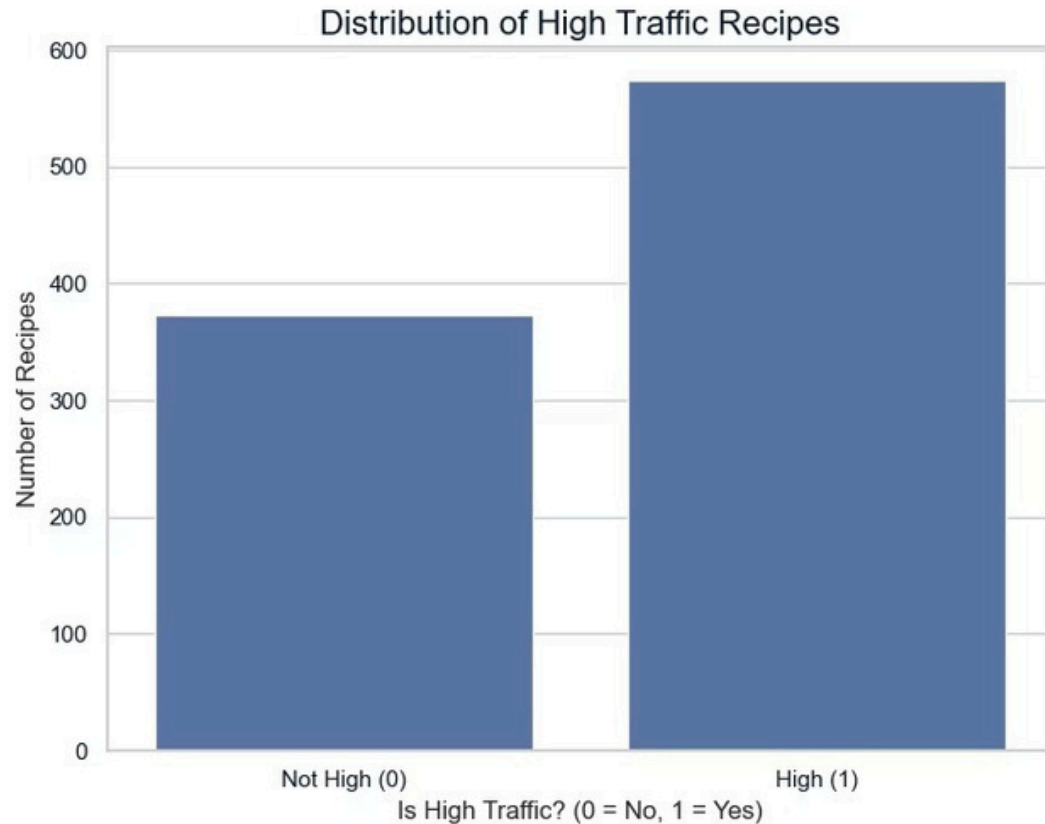
Analyzed which recipe features are linked to high traffic.

3

Predictive Modeling

Built and tested models to find the most accurate predictor.

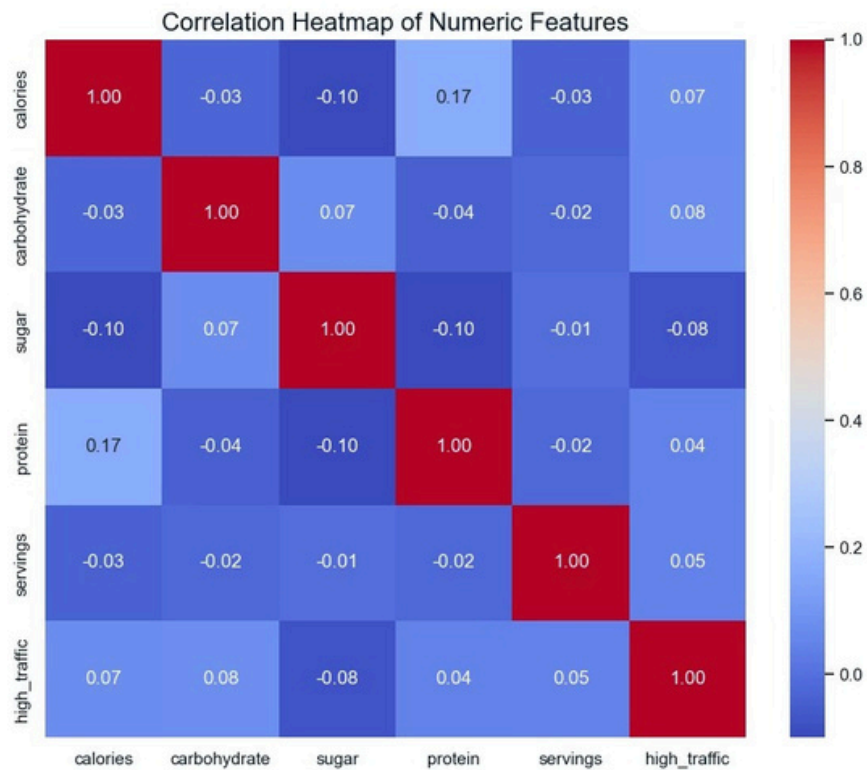
Understanding Our Data (Part 1: The Features)



Our features are complex: For example, 'calories' are heavily skewed, with many outliers. This means simple averages aren't reliable.

Our data is imbalanced: We have more "High" traffic recipes than "Not High." Our model had to be built to handle this.

Understanding Our Data (Part 2: The Relationships)



No single nutritional feature (like calories or protein) has a strong, simple relationship with high_traffic.

We needed a more sophisticated model to find the complex patterns.

Key Finding #1: Category is a Key Predictor

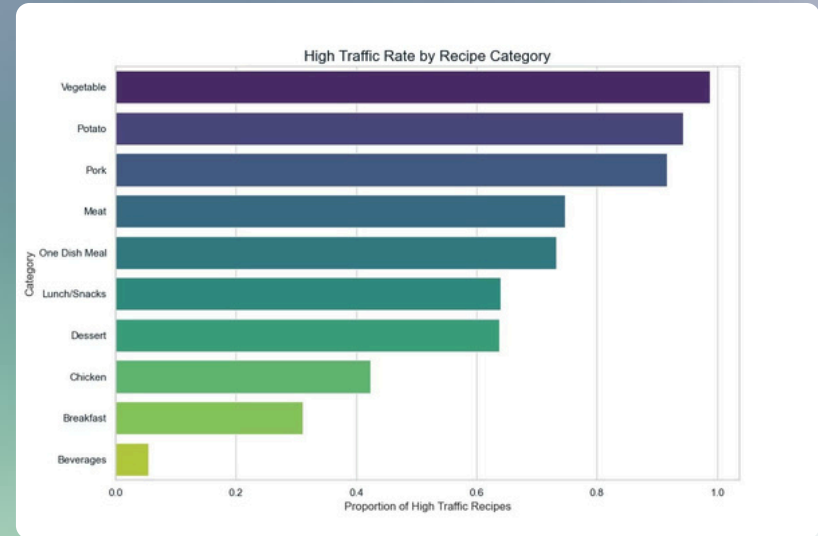
A recipe's category is a strong indicator of its potential for high traffic.

High Performers

Pork, Vegetable, and Potato recipes are consistently popular.

Low Performers

Beverages and Breakfast recipes are the least likely to drive high traffic.



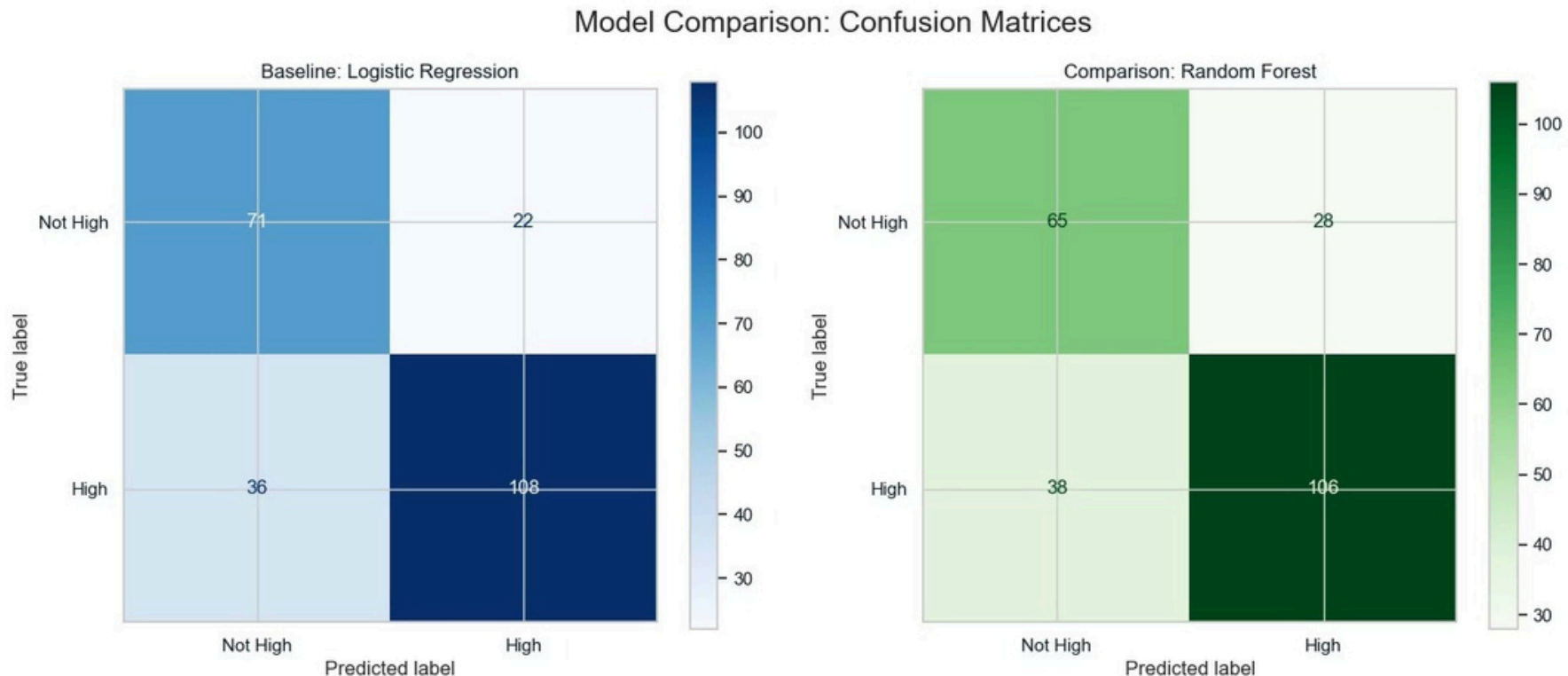
Key Finding #2: Building a Model That Works

We successfully built a predictive model (a Random Forest) that is highly effective at this task.

1. **Baseline Model (Left):** A simple model that struggled with the data's complexity.

2. **New Model (Right):** A more advanced model (Random Forest) that uses all the complex patterns we found.

Result: The new model is far more accurate at predicting both "High" and "Not High" traffic recipes.



Key Finding #3: We Met the Business Goal

The Business Metric: Your 80% target defines our metric: "High-Traffic Recall."

This answers: "Of all the recipes that would be popular, what percentage can our model find?"

Our Model's Performance

The new model (Random Forest) is a clear winner. It correctly identifies **84% of all high-traffic recipes**. It is significantly more accurate and reliable than the baseline model.

What This Means for You

→ No More
Guessing

You no longer have
to guess which recipe will
be popular.

→ Daily
Automation

We can now provide you with
a daily, automated
recommendation that is
statistically proven to meet
your
goal.

→ Measurable
Results

This allows us to more
reliably achieve the 40%
boost in site traffic and
drive more
subscriptions.

Recommendations & Next Steps

01

Implement the Model (Now)

Let's begin using this model's predictions to select the daily homepage recipe.

02

Monitor & Verify (Next Quarter)

We need to A/B test the model's picks to confirm and quantify the 40% traffic lift.

03

Improve (Future)

To make the model even smarter, we should gather more data like "Time to make" and "Cost per serving."

Thank You

Questions?